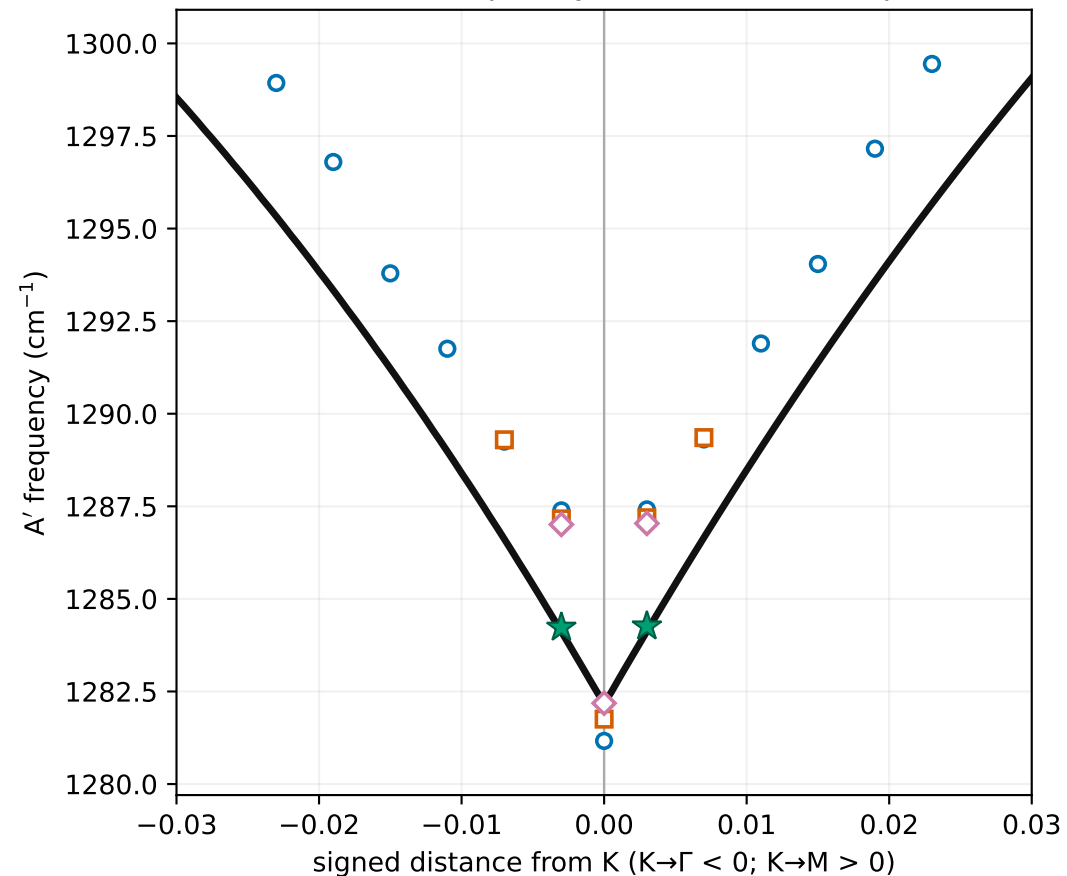
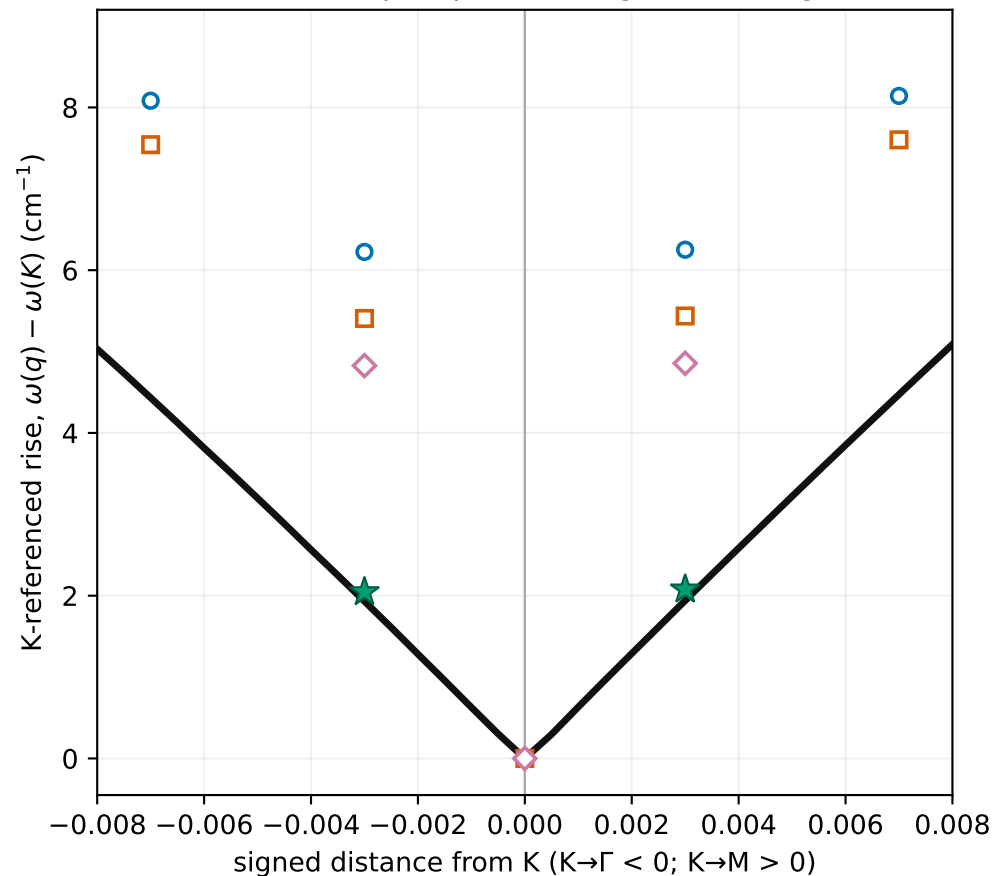


Graphene K-point A' at zero smearing: raw DFPT convergence and MLIP + long-range correction

(a) Absolute frequency and all raw DFPT points



(b) Near-K cusp depth and k-grid convergence



zero smearing/degauss (Ry)

— MLIP + full EPC LR

○ raw DFPT, k192 (13 points)

□ raw DFPT, k240 (5 points)

◇ raw DFPT, k288 (3 points)

★ DFPT $1/N_k$ depth limit at $|d| = 0.003$

DFPT integration: tetrahedra_opt; no degauss.
Raw markers are not joined or fitted.
The green stars extrapolate cusp depth only;
panel (a) anchors them to the k288 K frequency.

k240→k288 change in $d=0.003$ depth:
 $K \rightarrow \Gamma$ 12.00%; $K \rightarrow M$ 11.97%.